

ECHE 101

B.TECH. (IInd SEMESTER) EXAMINATION, 2025-26

BACHELOR OF TECHNOLOGY

(ME, CSE & CSE-AIML)

ENGINEERING CHEMISTRY

Time : Three Hours]

[Maximum Marks : 75

Note: There are **three** sections (A, B and C) and candidate has to attempt questions from all sections. Marks are indicated against each section.

Section-A

1. Answer **all** questions : 5×3=15

- (a) Distinguish between temporary and permanent hardness.
- (b) Define 'Blue Shift' and 'Red Shift' in UV spectroscopy.
- (c) Distinguish between Natural and synthetic polymers with suitable examples.
- (d) What is 'Phase Rule' ?
- (e) Define fuel and mention the characteristic of a good fuel.

Section-B

Note: Answer **all** questions of the following :

4×5=20

2. (a) Describe a galvanic cell with suitable diagram and Electrode reactions.

Or

- (b) Explain briefly the Zeolite process for water softening.

3. (a) In a polymer sample 30% molecule have molecular mass 40,000, 40% have molecular mass 30,000 and rest 30% have 60,000. Calculate the number average and weight average molecular mass.

Or

- (b) Discuss the structure and properties of Graphite.

4. (a) Draw the MO diagram of oxygen molecule. Arrange the following molecular species in their increasing order of stability (giving bond orders) : O_2 , O_2^+ , O_2^- , O_2^{2-}

Or

- (b) Discuss the preparation, properties and uses of 'SBR' and 'NBR'.

5. (a) Discuss the band structure of Insulators, conductors and semiconductors. Explain 'p-type' semiconductor with suitable diagram and example.

Or

- (b) Discuss the preparation, properties and uses of 'Neoprene' and 'Terylene'.

Section-C

Note: Answer any **two** questions of the following :

2×20=40

6. Define the terms phase, component and degree of freedom. With a neat sketch describe the phase diagram of water system and explain all the curves, equilibrium and point in the diagram.
7. What are nanomaterials ? Classify them on the basis of dimensions. Explain different approaches of synthesis and also mention their properties and application in various fields.

8. Write down the cell reaction. If Zn and Ag electrode are clipped in respected solution of their ions. Find out the E.M.F of the cell given that –

$$E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{v}$$

$$E_{\text{Ag}^+/\text{Ag}}^0 = 0.8\text{v}$$

Concentration of Zn^{2+} and Ag^+ ions are 0.2M and 0.01M respectively.

9. Write short note on ‘conducting polymers’ and ‘Biodegradable polymers’.

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